ModelIt!

Student Activity Pack

How Do Plants Grow from Seeds?

The Science of the Plant Life Cycle

Grade 1 | NGSS 1-LS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw what you think is inside a seed. Show what happens when you put it in soil and add water.

2 Circle what a seed needs to grow: water, candy, sunlight, soil, a hat, air.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Amount

Plant Growth

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Watered Every Day

Water Amount set to regular watering

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: No Water at All

Water Amount set to zero

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a tiny seed turn into a big plant? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design a Seed Starter Garden

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YOUR ENGINEERING MISSION

Build a mini garden that gives seeds the best chance to sprout and grow by providing water, light, and good soil.

THE SCENARIO

Your school wants to start a garden but needs to figure out the best way to grow seeds. Design a mini garden in a cup that gives your seed everything it needs to grow. Track how it changes each day!

Engineering Design Thinking

1. WHAT KIND OF CONTAINER WILL YOU USE FOR YOUR MINI GARDEN?
2. HOW MUCH WATER WILL YOU GIVE YOUR SEED EACH DAY?
3. WHERE WILL YOU PUT YOUR GARDEN SO IT GETS SUNLIGHT?
4. HOW WILL YOU RECORD YOUR PLANT'S GROWTH EACH DAY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Botanists study plants and help farmers grow healthier crops. They work in gardens, farms, and laboratories to discover how to grow better food. They earn $45,000-$85,000/year.